Elements of Information Theory **2026-2027**

Lecture 2: Method of types

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2.1 Types, type classes

Let $\mathcal{X} = \{a_1, \ldots, a_M\}$ be a finite alphabet, $M = |\mathcal{X}|$. The **type** of a sequence $x^n \in \mathcal{X}^n$ is its empirical distribution:

$$\hat{\mathrm{P}}_{x^n}(a) := \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{x_i = a\}, \qquad a \in \mathcal{X},$$
$$\hat{\mathrm{P}}_{x^n} := \big[\, \hat{\mathrm{P}}_{x^n}(a) : \ a \in \mathcal{X} \,\big].$$

A type is thus of the form $\hat{\mathrm{P}} = \frac{1}{n}\,[n_1, \ldots, n_M]$, where n_i is the number of occurrences of a_i in the sequence.

The **set of types** is

$$\begin{aligned}\mathcal{P}_n \equiv \mathcal{P}_n(\mathcal{X}) &:= \big\{\hat{\mathrm{P}}_{x^n} : \ x^n \in \mathcal{X}^n\big\} \\ &= \Big\{\frac{1}{n}\,[n_1, \ldots, n_M] : \ n_i \in \mathbb{Z}_+, \ \sum_{i=1}^{M} n_i = n\Big\} \\ &\subset \mathcal{P}(\mathcal{X}) := \Big\{[p_1, \ldots, p_M] : \ p_i \geq 0, \ \sum_{i=1}^{M} p_i = 1\Big\},\end{aligned}$$

where the set of distributions $\mathcal{P}(\mathcal{X})$ is the $(M-1)$-simplex.

The **type class** of a type $\hat{\mathrm{P}} \in \mathcal{P}_n$ is

$$\mathcal{T}^{(n)}(\hat{\mathrm{P}}) := \big\{x^n \in \mathcal{X}^n : \ \hat{\mathrm{P}}_{x^n} = \hat{\mathrm{P}}\big\}.$$

Example. Let $\mathcal{X} = \{0, 1\}$ (so $a_1 = 0$, $a_2 = 1$, $M = 2$) and $n = 4$. The sequence $x^4 = 0110$ contains $n_1 = 2$ zeros and $n_2 = 2$ ones, so its type is

$$\hat{\mathrm{P}}_{0110} = \frac{1}{4}\,[2, 2] = \Big[\frac{1}{2}, \frac{1}{2}\Big].$$

The set of types has $n + 1 = 5$ elements,

$$\mathcal{P}_4 = \Big\{[0, 1], \ \Big[\frac{1}{4}, \frac{3}{4}\Big], \ \Big[\frac{1}{2}, \frac{1}{2}\Big], \ \Big[\frac{3}{4}, \frac{1}{4}\Big], \ [1, 0]\Big\},$$

equally spaced points on the segment $\mathcal{P}(\mathcal{X}) = \big\{[p, 1-p] : \ p \in [0,1]\big\}$, which is the 1-simplex. The type class of $\big[\frac{1}{2}, \frac{1}{2}\big]$ consists of the $\binom{4}{2} = 6$ sequences with two zeros and two ones:

$$\mathcal{T}^{(4)}\Big(\Big[\frac{1}{2}, \frac{1}{2}\Big]\Big) = \{0011,\ 0101,\ 0110,\ 1001,\ 1010,\ 1100\}.$$

The five type classes, of sizes $1, 4, 6, 4, 1$, partition the $2^4 = 16$ sequences of $\mathcal{X}^4$.

In this lecture, we answer three questions:

- the number of types,
- the size of each type class,
- the probability of each type class.

2.2 Size and probability of type classes

2.2.1 Number of type classes

By "stars and bars" (Problem 2 of the exercises),

$$\begin{aligned} K_{n,M} := |\mathcal{P}_n| = \binom{n+M-1}{M-1} &\le (n+1)^{M-1} \\ &\le n^M \qquad \text{if } n \ge M \text{ and } n \ge 3. \end{aligned}$$

Proof. For the last bound, $(n+1)^{M-1} = n^{M-1}\big(1+\frac{1}{n}\big)^{M-1} \le n^{M-1}\big(1+\frac{1}{n}\big)^{n} < e\, n^{M-1} \le n^M$ when $M - 1 \le n$ and $n \ge 3$. □

2.2.2 Size of each type class

For a type $\hat{\mathrm{P}} \in \mathcal{P}_n$ (Problem 3 of the exercises),

$$\begin{aligned} |\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = \binom{n}{n_1, \ldots, n_M} &= \binom{n}{n\hat{\mathrm{P}}(a_1), \ldots, n\hat{\mathrm{P}}(a_M)} \\ &= \frac{n!}{\prod_{a\in\mathcal{X}} \big(n\hat{\mathrm{P}}(a)\big)!}. \end{aligned}$$

For $M = 2$, a type is $\hat{\mathrm{P}} = \frac{1}{n}[k, n-k]$ with $k \in \{0, \ldots, n\}$, and we get the binomial coefficient

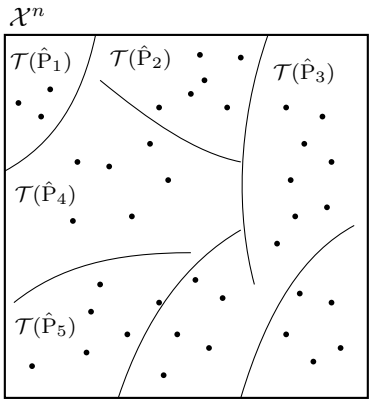
$$|\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = \binom{n}{k} = \frac{n!}{k!\,(n-k)!},$$

the number of binary sequences of length n with k occurrences of a_1. For example, $\binom{4}{2} = 6$ in the example of Section 2.1.

The type classes partition $\mathcal{X}^n$, as sketched below. Counting the sequences class by class therefore gives

$$\sum_{\hat{\mathrm{P}}\in\mathcal{P}_n} |\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = \sum_{\substack{n_1,\ldots,n_M \in \mathbb{Z}_+ \\ n_1+\cdots+n_M = n}} \binom{n}{n_1, \ldots, n_M} = M^n,$$

which is the multinomial theorem for $(1 + \cdots + 1)^n$. For $M = 2$, it reads $\sum_{k=0}^{n} \binom{n}{k} = 2^n$, e.g., $1 + 4 + 6 + 4 + 1 = 16$ for $n = 4$.



Stirling's approximation. For every integer $n \geq 1$ (Robbins' form, which we use without proof),

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n+\theta_n}}, \qquad \theta_n \in [0, 1].$$

Applied to each factorial, it gives, for $\hat{\mathrm{P}} \in \mathcal{P}_n$ with $n_i = n\hat{\mathrm{P}}(a_i) \geq 1$ for all i,

$$|\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = 2^{nH(\hat{\mathrm{P}})} \sqrt{\frac{n}{n_1 \cdots n_M} (2\pi)^{-(M-1)}}\, e^{\frac{1}{12n+\theta_n} - \sum_{i=1}^{M} \frac{1}{12n_i+\theta_{n_i}}}.$$

Indeed, since $n = \sum_i n_i$, the factors e^{-n} and $\prod_i e^{-n_i}$ cancel, and $n^n / \prod_i n_i^{n_i} = 2^{nH(\hat{\mathrm{P}})}$ since $n_i/n = \hat{\mathrm{P}}(a_i)$. The factor after $2^{nH(\hat{\mathrm{P}})}$ is polynomial in n; we come back to it in Section 2.2.3. If $n_i = 0$ for some i, we simply remove a_i from the alphabet: the formula above holds with M replaced by $\tilde{M} := |\{a \in \mathcal{X} : \hat{\mathrm{P}}(a) > 0\}|$, the effective size of the alphabet for the type $\hat{\mathrm{P}}$.

2.2.3 Probability of type classes

Let $\mathrm{Q} \in \mathcal{P}(\mathcal{X})$ be a probability measure with pmf q, and let Q^n be the i.i.d. distribution on $\mathcal{X}^n$, that is, the product measure with pmf $q^n(x^n) := \prod_{i=1}^{n} q(x_i)$. Since the alphabet is finite, a measure is determined by its pmf, and we use the same letter for both: $\mathrm{Q}(a) := \mathrm{Q}(\{a\}) = q(a)$, $\mathrm{Q}^n(x^n) := \mathrm{Q}^n(\{x^n\}) = q^n(x^n)$, and $\mathrm{Q}^n(\mathcal{A}) = \sum_{x^n \in \mathcal{A}} \mathrm{Q}^n(x^n)$ for every $\mathcal{A} \subseteq \mathcal{X}^n$. The same shorthand is used for P and for types.

- **Probability of a sequence under Q^n.**

$$\begin{aligned} \mathrm{Q}^n(x^n) &= \prod_{i=1}^{n} \mathrm{Q}(x_i) = \prod_{a \in \mathcal{X}} \mathrm{Q}(a)^{n\hat{\mathrm{P}}(a)} \\ &= 2^{n \sum_{a \in \mathcal{X}} \hat{\mathrm{P}}(a) \log \mathrm{Q}(a)} \\ &= 2^{-nH(\hat{\mathrm{P}} \| \mathrm{Q})}, \end{aligned}$$

  where $\hat{\mathrm{P}} := \hat{\mathrm{P}}_{x^n}$ is the type of x^n. Since $H(\hat{\mathrm{P}} \| \mathrm{Q}) = H(\hat{\mathrm{P}}) + D(\hat{\mathrm{P}} \| \mathrm{Q})$, with $D(\hat{\mathrm{P}} \| \mathrm{Q}) = 0$ if and only if $\mathrm{Q} = \hat{\mathrm{P}}$, the **maximum-likelihood** estimate of the source from x^n is its

type:

$$\mathrm{Q}^* := \arg\max_{\mathrm{Q}\in\mathcal{P}(\mathcal{X})} \mathrm{Q}^n(x^n) = \hat{\mathrm{P}}_{x^n}.$$

- **Probability of a type class.** For every $\hat{\mathrm{P}} \in \mathcal{P}_n$,

$$\begin{aligned}\mathrm{Q}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big) &= \sum_{x^n\in\mathcal{T}^{(n)}(\hat{\mathrm{P}})} \mathrm{Q}^n(x^n) \\ &= |\mathcal{T}^{(n)}(\hat{\mathrm{P}})|\, 2^{-nH(\hat{\mathrm{P}}\|\mathrm{Q})}. \qquad \circledast\end{aligned}$$

- **Most likely source for a type class.** For every $\hat{\mathrm{P}} \in \mathcal{P}_n$ and every $\mathrm{Q} \in \mathcal{P}(\mathcal{X})$,

$$\hat{\mathrm{P}}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big) \ge \mathrm{Q}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big),$$

from $\circledast$, since $H(\hat{\mathrm{P}}\|\mathrm{Q}) \ge H(\hat{\mathrm{P}})$.

- **Most likely type class under a source.** For every $\hat{\mathrm{P}}, \hat{\mathrm{Q}} \in \mathcal{P}_n$,

$$\hat{\mathrm{P}}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big) \ge \hat{\mathrm{P}}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{Q}})\big),$$

that is, the type class of $\hat{\mathrm{P}}$ is the most likely one under $\hat{\mathrm{P}}^n$. Indeed, using $s!/t! \ge t^{s-t}$ with $s = n\hat{\mathrm{Q}}(a)$ and $t = n\hat{\mathrm{P}}(a)$, we have

$$\frac{|\mathcal{T}^{(n)}(\hat{\mathrm{P}})|}{|\mathcal{T}^{(n)}(\hat{\mathrm{Q}})|} = \prod_{a\in\mathcal{X}} \frac{(n\hat{\mathrm{Q}}(a))!}{(n\hat{\mathrm{P}}(a))!} \ge \prod_{a\in\mathcal{X}} \big(n\hat{\mathrm{P}}(a)\big)^{n(\hat{\mathrm{Q}}(a)-\hat{\mathrm{P}}(a))} = 2^{n(H(\hat{\mathrm{P}})-H(\hat{\mathrm{Q}}\|\hat{\mathrm{P}}))},$$

so, by $\circledast$,

$$\begin{aligned}\frac{\hat{\mathrm{P}}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big)}{\hat{\mathrm{P}}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{Q}})\big)} &= \frac{|\mathcal{T}^{(n)}(\hat{\mathrm{P}})|}{|\mathcal{T}^{(n)}(\hat{\mathrm{Q}})|}\,\frac{2^{-nH(\hat{\mathrm{P}}\|\hat{\mathrm{P}})}}{2^{-nH(\hat{\mathrm{Q}}\|\hat{\mathrm{P}})}} \\ &\ge 2^{n(H(\hat{\mathrm{P}})-H(\hat{\mathrm{Q}}\|\hat{\mathrm{P}}))-n(H(\hat{\mathrm{P}})-H(\hat{\mathrm{Q}}\|\hat{\mathrm{P}}))} \\ &= 1.\end{aligned}$$

If the uniform distribution $\hat{\mathrm{P}}$ is a type (M divides n), then $H(\hat{\mathrm{Q}}\|\hat{\mathrm{P}}) = \log M = H(\hat{\mathrm{P}})$ for every $\hat{\mathrm{Q}}$, so the size bound above has exponent 0: the uniform type class is the largest.

- **Maximum-likelihood probability of a type class.** For $\hat{\mathrm{P}} \in \mathcal{P}_n$, it is defined as

$$\gamma_n(\hat{\mathrm{P}}) := \hat{\mathrm{P}}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big) = \max_{\mathrm{Q}\in\mathcal{P}(\mathcal{X})} \mathrm{Q}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big),$$

since the maximum is attained at $\mathrm{Q} = \hat{\mathrm{P}}$, as shown above. By $\circledast$ with $\mathrm{Q} = \hat{\mathrm{P}}$,

$$|\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = 2^{nH(\hat{\mathrm{P}})}\,\gamma_n(\hat{\mathrm{P}}).$$

Since the $K_{n,M}$ type classes partition $\mathcal{X}^n$ and $\mathcal{T}^{(n)}(\hat{\mathrm{P}})$ is the most likely one under $\hat{\mathrm{P}}^n$, we have $1 \ge \gamma_n(\hat{\mathrm{P}}) \ge K_{n,M}^{-1}$. Hence, with $\eta_n := \frac{1}{n}\log\gamma_n(\hat{\mathrm{P}})$,

$$|\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = 2^{n(H(\hat{\mathrm{P}})+\eta_n)}, \qquad -\frac{\log K_{n,M}}{n} \le \eta_n \le 0,$$

so that $\eta_n \to 0$ as $n \to \infty$, uniformly over $\hat{\mathrm{P}} \in \mathcal{P}_n$. Moreover, Stirling's approximation in Section 2.2.2 gives (Problem 4 of the exercises)

$$\gamma_n(\hat{\mathrm{P}}) \simeq (2\pi n)^{-\frac{\tilde{M}-1}{2}}\, \tilde{M}^{\frac{\tilde{M}}{2}}\, 2^{\frac{\tilde{M}}{2} D(\mathrm{U}\|\hat{\mathrm{P}})},$$

where U is the uniform distribution on $\mathrm{supp}(\hat{\mathrm{P}})$, $\tilde{M}$ is the effective alphabet size defined in Section 2.2.2, and $\simeq$ means that the ratio of both sides tends to 1 as $n \to \infty$. So $\gamma_n(\hat{\mathrm{P}})$ is polynomial in n, and goes to 0 polynomially fast if $\tilde{M} \geq 2$! Among types with the same support size $\tilde{M}$, the uniform type has the smallest maximum-likelihood probability, although its type class is the largest.

- For $\hat{\mathrm{P}} \in \mathcal{P}_n$ and $\mathrm{Q} \in \mathcal{P}(\mathcal{X})$,

$$\begin{aligned} \mathrm{Q}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big) &= |\mathcal{T}^{(n)}(\hat{\mathrm{P}})|\, 2^{-nH(\hat{\mathrm{P}}\|\mathrm{Q})} \\ &= \gamma_n(\hat{\mathrm{P}})\, 2^{nH(\hat{\mathrm{P}})-nH(\hat{\mathrm{P}}\|\mathrm{Q})} \\ &= \gamma_n(\hat{\mathrm{P}})\, 2^{-nD(\hat{\mathrm{P}}\|\mathrm{Q})}. \end{aligned}$$

  If $\hat{\mathrm{P}} \neq \mathrm{Q}$, then $D(\hat{\mathrm{P}}\|\mathrm{Q}) > 0$, and the type class of $\hat{\mathrm{P}}$ has an exponentially low probability under Q^n.

Summary. For $\hat{\mathrm{P}} \in \mathcal{P}_n$, $x^n \in \mathcal{T}^{(n)}(\hat{\mathrm{P}})$, and $\mathrm{Q} \in \mathcal{P}(\mathcal{X})$,

$$|\mathcal{P}_n| = K_{n,M} \leq (n+1)^{M-1}, \qquad \mathrm{Q}^n(x^n) = 2^{-nH(\hat{\mathrm{P}}\|\mathrm{Q})},$$
$$|\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = \gamma_n(\hat{\mathrm{P}})\, 2^{nH(\hat{\mathrm{P}})}, \qquad \mathrm{Q}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big) = \gamma_n(\hat{\mathrm{P}})\, 2^{-nD(\hat{\mathrm{P}}\|\mathrm{Q})},$$

where $\gamma_n(\hat{\mathrm{P}}) := \hat{\mathrm{P}}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big) \in \big[K_{n,M}^{-1}, 1\big]$.

2.3 Typical sequences

Let $\mathrm{P} \in \mathcal{P}(\mathcal{X})$ and $\varepsilon \in [0,1]$. A sequence $x^n \in \mathcal{X}^n$ is called ε**-typical** with respect to P, denoted by $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$, if its type is close to P:

$$|\hat{\mathrm{P}}_{x^n}(a) - \mathrm{P}(a)| \leq \varepsilon\, \mathrm{P}(a), \quad \text{i.e.,} \quad (1-\varepsilon)\mathrm{P}(a) \leq \hat{\mathrm{P}}_{x^n}(a) \leq (1+\varepsilon)\mathrm{P}(a), \qquad \forall\, a \in \mathcal{X}.$$

Therefore, we call

$$\begin{aligned} \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}) &:= \big\{x^n :\ |\hat{\mathrm{P}}_{x^n}(a) - \mathrm{P}(a)| \leq \varepsilon\, \mathrm{P}(a),\ \forall\, a \in \mathcal{X}\big\} \\ &= \bigcup \big\{\mathcal{T}^{(n)}(\hat{\mathrm{P}}) :\ \hat{\mathrm{P}} \in \mathcal{P}_n,\ |\hat{\mathrm{P}}(a) - \mathrm{P}(a)| \leq \varepsilon\, \mathrm{P}(a),\ \forall\, a \in \mathcal{X}\big\} \end{aligned}$$

the **strongly typical set**, or set of ε-typical n-sequences; it is a union of type classes. If $\mathrm{P}(a) = 0$, the constraint forces $\hat{\mathrm{P}}_{x^n}(a) = 0$, so the type of a typical sequence is absolutely continuous with respect to P: $\hat{\mathrm{P}}_{x^n} \ll \mathrm{P}$ for every $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$. If $\varepsilon < 1$, then $\hat{\mathrm{P}}_{x^n}(a) \geq (1-\varepsilon)\mathrm{P}(a) > 0$ whenever $\mathrm{P}(a) > 0$, so the two even have the same support. For $\varepsilon = 0$, the typical set is $\mathcal{T}^{(n)}(\mathrm{P})$ if P is a type, and $\emptyset$ otherwise. Typicality can also be written as

$$\Big|\frac{\hat{\mathrm{P}}_{x^n}(a)}{\mathrm{P}(a)} - 1\Big| \leq \varepsilon, \qquad \forall\, a \in \mathcal{X},$$

with the conventions $c/0 = \infty$ for $c > 0$ and $0/0 = 1$.

Typical average lemma. For every function $g : \mathcal{X} \to \mathbb{R}$ and every $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$,

$$\Big|\frac{1}{n}\sum_{i=1}^{n} g(x_i) - \mathbb{E}_{\mathrm{P}}\big[g(\mathrm{X})\big]\Big| \le \varepsilon\, \mathbb{E}_{\mathrm{P}}\big[|g(\mathrm{X})|\big].$$

Proof. The average is $\mathbb{E}_{\hat{\mathrm{P}}_{x^n}}[g(\mathrm{X})]$, and $|\hat{\mathrm{P}}_{x^n}(a)/\mathrm{P}(a) - 1| \le \varepsilon$ for all a. Since $\hat{\mathrm{P}}_{x^n} \ll \mathrm{P}$, the ratio $\hat{\mathrm{P}}_{x^n}(a)/\mathrm{P}(a)$ is the Radon–Nikodym derivative $\frac{\mathrm{d}\hat{\mathrm{P}}_{x^n}}{\mathrm{d}\mathrm{P}}(a)$, and by a change of measure,

$$\Big|\mathbb{E}_{\hat{\mathrm{P}}_{x^n}}\big[g(\mathrm{X})\big] - \mathbb{E}_{\mathrm{P}}\big[g(\mathrm{X})\big]\Big| = \Big|\mathbb{E}_{\mathrm{P}}\Big[\Big(\frac{\hat{\mathrm{P}}_{x^n}(\mathrm{X})}{\mathrm{P}(\mathrm{X})} - 1\Big)g(\mathrm{X})\Big]\Big| \le \varepsilon\, \mathbb{E}_{\mathrm{P}}\big[|g(\mathrm{X})|\big]. \qquad \square$$

Properties. Let $M^* := |\{a \in \mathcal{X} : \ \mathrm{P}(a) > 0\}|$, $c_{\mathrm{P}} := \min_{a:\,\mathrm{P}(a)>0} \mathrm{P}(a)^2$, and $\mathrm{X}^n \sim \mathrm{P}^n$.

1. **Probability of a sequence.** For every $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$,

   $$2^{-n(1+\varepsilon)H(\mathrm{P})} \le \mathrm{P}^n(x^n) \le 2^{-n(1-\varepsilon)H(\mathrm{P})},$$

   $D(\hat{\mathrm{P}}_{x^n}\|\mathrm{P}) \le \log(1+\varepsilon)$, and $|H(\hat{\mathrm{P}}_{x^n}) - H(\mathrm{P})| \le \varepsilon H(\mathrm{P}) + \log(1+\varepsilon)$.

2. **Probability of the set.**

   $$\mathrm{P}^n\big(\mathrm{X}^n \notin \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})\big) \le \delta_T(\varepsilon, n, \mathrm{P}) := 2M^* e^{-2n\varepsilon^2 c_{\mathrm{P}}},$$

   so $\mathrm{P}^n\big(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})\big) \to 1$ if $n\varepsilon^2 \to \infty$.

3. **Size of the set.**

   $$\big(1 - \delta_T(\varepsilon, n, \mathrm{P})\big)\, 2^{n(1-\varepsilon)H(\mathrm{P})} \le |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})| \le 2^{n(1+\varepsilon)H(\mathrm{P})}.$$

Proof. 1. Apply the lemma to $g := \log \mathrm{P}$: we have $\frac{1}{n}\log \mathrm{P}^n(x^n) = \frac{1}{n}\sum_{i=1}^n \log \mathrm{P}(x_i)$ and $\mathbb{E}_{\mathrm{P}}\big[|\log \mathrm{P}(\mathrm{X})|\big] = H(\mathrm{P})$. Next, $\log \frac{\hat{\mathrm{P}}_{x^n}(a)}{\mathrm{P}(a)} \le \log(1+\varepsilon)$ whenever $\hat{\mathrm{P}}_{x^n}(a) > 0$, and averaging under $\hat{\mathrm{P}}_{x^n}$ bounds the divergence. Finally, $H(\hat{\mathrm{P}}_{x^n}) = H(\hat{\mathrm{P}}_{x^n}\|\mathrm{P}) - D(\hat{\mathrm{P}}_{x^n}\|\mathrm{P})$, and the lemma with $g = \log \mathrm{P}$ also gives $|H(\hat{\mathrm{P}}_{x^n}\|\mathrm{P}) - H(\mathrm{P})| \le \varepsilon H(\mathrm{P})$.

We write the bounds on $\mathrm{P}^n(x^n)$ as $\mathrm{P}^n(x^n) \approx 2^{-nH(\mathrm{P})}$. Sequences satisfying them are called **weakly typical**, so strong typicality implies weak typicality. The converse fails: if P is uniform and $M \ge 2$, every x^n is weakly typical, including $x^n = a_1 a_1 \cdots a_1$, which is not strongly typical for $\varepsilon < 1$.

2. For each $a \in \mathcal{X}$, the random variables $\mathbf{1}\{\mathrm{X}_1 = a\}, \ldots, \mathbf{1}\{\mathrm{X}_n = a\}$ are i.i.d., with values in $[0,1]$ and mean $\mathrm{P}(a)$, and their arithmetic mean is $\hat{\mathrm{P}}_{\mathrm{X}^n}(a)$. If $\mathrm{P}(a) = 0$, then $\hat{\mathrm{P}}_{\mathrm{X}^n}(a) = 0$ with probability 1. By the union bound and **Hoeffding's inequality** (the mean of n independent random variables with values in $[0,1]$ deviates from its

expectation by more than t with probability at most $2e^{-2nt^2}$),

$$\begin{aligned}\mathrm{P}^n\big(\mathrm{X}^n \notin \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})\big) &\le \sum_{a:\,\mathrm{P}(a)>0} \mathrm{P}^n\Big(\Big|\frac{1}{n}\sum_{i=1}^n \mathbf{1}\{\mathrm{X}_i = a\} - \mathrm{P}(a)\Big| > \varepsilon \mathrm{P}(a)\Big) \\ &\le \sum_{a:\,\mathrm{P}(a)>0} 2e^{-2n\varepsilon^2 \mathrm{P}(a)^2} \le 2M^* e^{-2n\varepsilon^2 c_{\mathrm{P}}}.\end{aligned}$$

3. By item 1, $|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})|\, 2^{-n(1+\varepsilon)H(\mathrm{P})} \le \mathrm{P}^n\big(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})\big) \le |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})|\, 2^{-n(1-\varepsilon)H(\mathrm{P})}$. Use $\mathrm{P}^n\big(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})\big) \le 1$ on the left and $\mathrm{P}^n\big(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})\big) \ge 1 - \delta_T(\varepsilon, n, \mathrm{P})$ (item 2) on the right. □

So the probability of an i.i.d. source P^n concentrates on $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$, a set of about $2^{nH(\mathrm{P})}$ sequences out of all $M^n = 2^{n\log M}$, and every sequence in it has probability about $2^{-nH(\mathrm{P})}$. This is the **asymptotic equipartition property** (AEP).

Comparison with type classes. The properties above parallel those of a type class (Section 2.2.3), with probabilities under $\hat{\mathrm{P}}^n$ for the type class $\mathcal{T}^{(n)}(\hat{\mathrm{P}})$ and under P^n for the typical set $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$:

	type class $\mathcal{T}^{(n)}(\hat{\mathrm{P}})$	typical set $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$
probability of a sequence	$2^{-nH(\hat{\mathrm{P}})}$	$2^{-n(1\pm\varepsilon)H(\mathrm{P})}$
size of the set	$\gamma_n(\hat{\mathrm{P}})\, 2^{nH(\hat{\mathrm{P}})}$	$2^{n(1\pm\varepsilon)H(\mathrm{P})}$
probability of the set	$\gamma_n(\hat{\mathrm{P}}) \to 0$ polynomially	$\ge 1 - \delta_T(\varepsilon, n, \mathrm{P}) \to 1$

where the exponents $(1\pm\varepsilon)H(\mathrm{P})$ stand for the bounds of items 1 and 3. A single type class has a vanishing probability, but only polynomially fast: $\gamma_n(\hat{\mathrm{P}}) \ge K_{n,M}^{-1} \ge (n+1)^{-(M-1)}$, so the loss is invisible on the exponential scale. In contrast, $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$, a union of the type classes close to P, carries almost all the probability; the price is the slack $\pm\varepsilon$ in the exponents.

Conversely, type classes give a tighter lower bound on the size, without any concentration argument. For every type $\hat{\mathrm{P}} \in \mathcal{P}_n$ with $|\hat{\mathrm{P}}(a) - \mathrm{P}(a)| \le \varepsilon \mathrm{P}(a)$ for all a,

$$|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})| \ge |\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = \gamma_n(\hat{\mathrm{P}})\, 2^{nH(\hat{\mathrm{P}})} \ge K_{n,M}^{-1}\, 2^{nH(\hat{\mathrm{P}})}.$$

Such a type exists for n large enough. Rounding each $n\mathrm{P}(a)$ up or down gives $\hat{\mathrm{P}} \in \mathcal{P}_n$ with $|\hat{\mathrm{P}}(a) - \mathrm{P}(a)| < \frac{1}{n}$ for all a, and $\hat{\mathrm{P}}(a) = 0$ if $\mathrm{P}(a) = 0$. With $\mathrm{P}_{\min} := \min_{a:\,\mathrm{P}(a)>0} \mathrm{P}(a)$ and $\tau_n := \frac{1}{n\mathrm{P}_{\min}}$, we get $|\hat{\mathrm{P}}(a) - \mathrm{P}(a)| \le \tau_n \mathrm{P}(a)$ for all a, so item 1, applied with τ_n to any $x^n \in \mathcal{T}^{(n)}(\hat{\mathrm{P}})$, gives

$$H(\hat{\mathrm{P}}) \ge (1-\tau_n)H(\mathrm{P}) - \log(1+\tau_n) \ge H(\mathrm{P}) - \frac{H(\mathrm{P}) + \log e}{n\mathrm{P}_{\min}},$$

since $\log(1+\tau_n) \le \tau_n \log e$. Hence, for $n \ge \frac{1}{\varepsilon \mathrm{P}_{\min}}$ (so that $\tau_n \le \varepsilon$),

$$|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})| \ge K_{n,M}^{-1}\, 2^{nH(\mathrm{P}) - \frac{H(\mathrm{P})+\log e}{\mathrm{P}_{\min}}},$$

that is, $\frac{1}{n}\log|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})| \ge H(\mathrm{P}) - O\big(\frac{\log n}{n}\big)$, much better than $(1-\varepsilon)H(\mathrm{P})$.

2.4 Jointly and conditionally typical sequences

Let $\mathrm{P}_{\mathrm{XY}} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ be a joint distribution with marginals P_{X} and P_{Y}. In the same way, the **jointly typical set** is

$$\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}}) := \Big\{(x^n, y^n) : \ \big|\hat{\mathrm{P}}_{x^n,y^n}(a,b) - \mathrm{P}_{\mathrm{XY}}(a,b)\big| \le \varepsilon\, \mathrm{P}_{\mathrm{XY}}(a,b),\ \forall\,(a,b) \in \mathcal{X} \times \mathcal{Y}\Big\},$$

and, for $y^n \in \mathcal{Y}^n$, the **conditionally typical set** is

$$\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}} \,|\, y^n) := \big\{x^n : \ (x^n, y^n) \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}})\big\}.$$

By definition,

$$(x^n, y^n) \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}}) \iff x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}} \,|\, y^n) \iff y^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}} \,|\, x^n).$$

Below, $\mathrm{P}^n(x^n) \approx 2^{-n(1\pm\varepsilon)H}$ means $2^{-n(1+\varepsilon)H} \le \mathrm{P}^n(x^n) \le 2^{-n(1-\varepsilon)H}$, and δ_T is defined in Section 2.3.

Properties.

1. If $(x^n, y^n) \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}})$, then $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{X}})$, $y^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{Y}})$,
$$\mathrm{P}_{\mathrm{X}}^n(x^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{X})}, \quad \mathrm{P}_{\mathrm{Y}}^n(y^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{Y})},$$
$$\mathrm{P}_{\mathrm{XY}}^n(x^n, y^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{X},\mathrm{Y})},$$
$$\mathrm{P}_{\mathrm{Y|X}}^n(y^n \,|\, x^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{Y}\,|\,\mathrm{X})}, \quad \mathrm{P}_{\mathrm{X|Y}}^n(x^n \,|\, y^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{X}\,|\,\mathrm{Y})}.$$
2. $\big(1 - \delta_T(\varepsilon, n, \mathrm{P}_{\mathrm{XY}})\big)\, 2^{n(1-\varepsilon)H(\mathrm{X},\mathrm{Y})} \le |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}})| \le 2^{n(1+\varepsilon)H(\mathrm{X},\mathrm{Y})}$.
3. For every $y^n \in \mathcal{Y}^n$, $|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}} \,|\, y^n)| \le 2^{n(1+\varepsilon)H(\mathrm{X}\,|\,\mathrm{Y})}$.
4. **Conditional typicality lemma.** Let $\mathrm{P}_{\mathrm{X}^n|\mathrm{Y}^n} = \mathrm{P}_{\mathrm{X|Y}}^n$. For every $y^n \in \mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_{\mathrm{Y}})$,
$$\mathrm{P}_{\mathrm{X}^n|\mathrm{Y}^n=y^n}\big(\mathrm{X}^n \notin \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}} \,|\, y^n)\big) \le \delta_T(\varepsilon/2, n, \mathrm{P}_{\mathrm{XY}}),$$
which tends to 0 if $n\varepsilon^2 \to \infty$.
5. For every $y^n \in \mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_{\mathrm{Y}})$,
$$|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}} \,|\, y^n)| \ge \big(1 - \delta_T(\varepsilon/2, n, \mathrm{P}_{\mathrm{XY}})\big)\, 2^{n(1-\varepsilon)H(\mathrm{X}\,|\,\mathrm{Y})}.$$
6. **Joint typicality lemma.** For every $y^n \in \mathcal{Y}^n$,
$$\mathrm{P}_{\mathrm{X}}^n\big(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}} \,|\, y^n)\big) \le 2^{-n(I(\mathrm{X};\mathrm{Y})-2\varepsilon H(\mathrm{X}))},$$
and for every $y^n \in \mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_{\mathrm{Y}})$,
$$\mathrm{P}_{\mathrm{X}}^n\big(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{\mathrm{XY}} \,|\, y^n)\big) \ge \big(1 - \delta_T(\varepsilon/2, n, \mathrm{P}_{\mathrm{XY}})\big)\, 2^{-n(I(\mathrm{X};\mathrm{Y})+2\varepsilon H(\mathrm{X}))}.$$
So an $\mathrm{X}^n \sim \mathrm{P}_{\mathrm{X}}^n$ drawn independently of y^n is jointly typical with y^n with probability about $2^{-nI(\mathrm{X};\mathrm{Y})}$.

Proof. 1. Summing over b, $\hat{\mathrm{P}}_{x^n}(a) = \sum_b \hat{\mathrm{P}}_{x^n,y^n}(a,b)$ is within $\varepsilon \mathrm{P_X}(a)$ of $\mathrm{P_X}(a)$, and similarly for y^n. The bounds follow from the typical average lemma for $\mathrm{P_{XY}}$ with $g(a,b) = \log \mathrm{P_X}(a)$, $\log \mathrm{P_Y}(b)$, $\log \mathrm{P_{XY}}(a,b)$, $\log \mathrm{P_{Y|X}}(b \,|\, a)$ and $\log \mathrm{P_{X|Y}}(a \,|\, b)$. For example, $\log \mathrm{P}^n_{\mathrm{X|Y}}(x^n \,|\, y^n) = \sum_{i=1}^n \log \mathrm{P_{X|Y}}(x_i \,|\, y_i)$ and $\mathbb{E}\big[\,|\log \mathrm{P_{X|Y}}(\mathrm{X} \,|\, \mathrm{Y})|\big] = H(\mathrm{X} \,|\, \mathrm{Y})$.

2. This is property 3 of Section 2.3 applied to $\mathrm{P_{XY}}$.

3. By item 1,

$$1 \geq \sum_{x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P_{XY}} \,|\, y^n)} \mathrm{P}^n_{\mathrm{X|Y}}(x^n \,|\, y^n) \geq |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P_{XY}} \,|\, y^n)|\, 2^{-n(1+\varepsilon)H(\mathrm{X} \,|\, \mathrm{Y})}.$$

4. Given $\mathrm{Y}^n = y^n$, the X_i are independent with $\mathrm{X}_i \sim \mathrm{P_{X|Y=}}_{y_i}$. If $\mathrm{P_{XY}}(a,b) = 0$, then $\hat{\mathrm{P}}_{\mathrm{X}^n,y^n}(a,b) = 0$ with probability 1. Otherwise, $\hat{\mathrm{P}}_{\mathrm{X}^n,y^n}(a,b)$ is the mean of the n independent random variables $\mathbf{1}\{(\mathrm{X}_i, y_i) = (a,b)\}$, with values in $[0,1]$, and its expectation is $\hat{\mathrm{P}}_{y^n}(b)\mathrm{P_{X|Y}}(a \,|\, b)$. Since $y^n \in \mathcal{T}^{(n)}_{\varepsilon/2}(\mathrm{P_Y})$, this expectation is within $\frac{\varepsilon}{2}\mathrm{P_{XY}}(a,b)$ of $\mathrm{P_{XY}}(a,b)$. A deviation from $\mathrm{P_{XY}}(a,b)$ by more than $\varepsilon \mathrm{P_{XY}}(a,b)$ therefore needs a deviation from the expectation by more than $\frac{\varepsilon}{2}\mathrm{P_{XY}}(a,b)$, which has probability at most $2e^{-n\varepsilon^2 \mathrm{P_{XY}}(a,b)^2/2}$ by Hoeffding's inequality. The union bound over the pairs with $\mathrm{P_{XY}}(a,b) > 0$ gives $\delta_T(\varepsilon/2, n, \mathrm{P_{XY}})$. (With y^n only in $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P_Y})$, the expectation can lie on the boundary, and the probability need not tend to 1.)

5. By item 4 and item 1,

$$\begin{aligned} 1 - \delta_T(\varepsilon/2, n, \mathrm{P_{XY}}) &\leq \mathrm{P}_{\mathrm{X}^n|\mathrm{Y}^n = y^n}\big(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P_{XY}} \,|\, y^n)\big) \\ &\leq |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P_{XY}} \,|\, y^n)|\, 2^{-n(1-\varepsilon)H(\mathrm{X} \,|\, \mathrm{Y})}. \end{aligned}$$

6. Every $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P_{XY}} \,|\, y^n)$ is in $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P_X})$, so $\mathrm{P}^n_{\mathrm{X}}(x^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{X})}$. With item 3,

$$\mathrm{P}^n_{\mathrm{X}}\big(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P_{XY}} \,|\, y^n)\big) \leq 2^{n(1+\varepsilon)H(\mathrm{X} \,|\, \mathrm{Y})}\, 2^{-n(1-\varepsilon)H(\mathrm{X})} \leq 2^{-n(I(\mathrm{X};\mathrm{Y}) - 2\varepsilon H(\mathrm{X}))},$$

since $H(\mathrm{X} \,|\, \mathrm{Y}) \leq H(\mathrm{X})$. The lower bound follows in the same way from item 5. □

Summary

- The type $\hat{\mathrm{P}}_{x^n}$ of a sequence is its empirical distribution. There are only $K_{n,M} \leq (n+1)^{M-1}$ types, polynomially many, for M^n sequences.
- All sequences of a type class have the same probability under an i.i.d. source: $\mathrm{Q}^n(x^n) = 2^{-nH(\hat{\mathrm{P}}\|\mathrm{Q})}$ for $x^n \in \mathcal{T}^{(n)}(\hat{\mathrm{P}})$. The most likely source for x^n is its type.
- A type class has $|\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = \gamma_n(\hat{\mathrm{P}})\, 2^{nH(\hat{\mathrm{P}})}$ sequences and probability $\mathrm{Q}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big) = \gamma_n(\hat{\mathrm{P}})\, 2^{-nD(\hat{\mathrm{P}}\|\mathrm{Q})}$, where $\gamma_n(\hat{\mathrm{P}}) \in \big[K_{n,M}^{-1}, 1\big]$ is polynomial in n. On the exponential scale, the size is 2^{nH} and the probability is 2^{-nD}.
- The typical set $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$ is the union of the type classes close to P. It has probability at least $1 - \delta_T(\varepsilon, n, \mathrm{P}) \to 1$ and about $2^{nH(\mathrm{P})}$ sequences, each of probability about $2^{-nH(\mathrm{P})}$ (AEP).
- An $\mathrm{X}^n \sim \mathrm{P}^n_{\mathrm{X}}$ drawn independently of a typical y^n is jointly typical with it with probability about $2^{-nI(\mathrm{X};\mathrm{Y})}$.